

1. ϕ -momentum:

$$\begin{aligned}
 & \rho \left(\frac{\partial v_\phi}{\partial t} + v_r \frac{\partial v_\phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\phi}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_\phi v_r + v_\theta v_\phi \cot \theta}{r} \right) \\
 &= -\frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} + \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\phi \sin \theta) \right) + \frac{1}{r \sin^2 \theta} \frac{\partial^2 v_\phi}{\partial \phi^2} + \frac{2}{r^2 \sin \theta} \frac{\partial v_r}{\partial \phi} \right. \\
 & \quad \left. + \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right] + \rho \frac{\partial v_\phi}{\partial t}
 \end{aligned}$$

Since $v_\phi(\theta)$, $v_r = 0$, $v_\theta = 0$

$$\Rightarrow 0 = \frac{\mu}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\phi \sin \theta) \right)$$

$$\frac{d}{d\theta} \left(\frac{1}{\sin \theta} \frac{d}{d\theta} (v_\phi \sin \theta) \right) = 0$$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} (v_\phi \sin \theta) = C_1$$

$$\frac{d}{d\theta} v_\phi = C_1$$

$$v_\phi = C_1 \theta + C_2$$

BCs: ① at $\theta = \frac{\pi}{2}$, $v_\phi = 0$

$$0 = C_1 \frac{\pi}{2} + C_2$$

② at $\theta = \frac{\pi}{2} - \theta_0$, $v_\phi = \Omega r$

$$\Omega r = C_1 \left(\frac{\pi}{2} - \theta_0 \right) + C_2$$

②-①:

$$\Omega r = -C_1 \theta_0$$

$$C_1 = -\frac{\Omega r}{\theta_0}$$

$$0 = -\frac{\Omega r}{\theta_0} \cdot \frac{\pi}{2} + C_2$$

$$C_2 = \frac{\pi}{2\theta_0} \Omega r$$

$$v_\phi = -\frac{\Omega r}{\theta_0} \theta + \frac{\pi}{2\theta_0} \Omega r = \Omega r \left(\frac{\pi}{2} - \frac{\theta}{\theta_0} \right)$$

$$\tau_{\theta\phi} = -\mu \left[\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{v_\phi}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial^2 v_\phi}{\partial \phi^2} \right]$$

$$= -\mu \left[\frac{1}{r} \frac{\partial}{\partial \theta} \left(\Omega r \left(\frac{\pi}{2} - \frac{\theta}{\theta_0} \right) \right) \right]$$

$$= -\mu \cdot \frac{1}{r} \cdot \Omega r \cdot \frac{-1}{\theta_0} = \frac{\mu \Omega}{\theta_0}$$

$$\Rightarrow T_z = F \cdot L = \tau_{\theta\phi} \cdot A \cdot r$$

$$= \int_0^{2\pi} \int_0^R (\tau_{\theta\phi} \cdot r) \cdot r dr d\phi$$

$$= \int_0^{2\pi} \int_0^R \frac{\mu \Omega}{\theta_0} \cdot r^2 dr d\phi$$

$$= \frac{\mu \Omega}{\theta_0} \cdot \frac{r^3}{3} \Big|_0^R \cdot \phi \Big|_0^{2\pi}$$

$$= \frac{\mu \Omega}{\theta_0} \cdot \frac{R^3}{3} \cdot 2\pi$$

$$= \frac{2\pi \mu \Omega}{3 \theta_0} R^3$$

$$\Rightarrow \mu = \frac{3}{2\pi} \frac{T_z \theta_0}{\Omega R^3}$$

2. Assume constant density and viscosity. (isothermal)

θ - momentum:

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta$$

$$\Rightarrow 0 = \mu \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right)$$

$$\frac{1}{r} \frac{d}{dr} (r v_\theta) = C_1$$

$$\frac{d}{dr} (r v_\theta) = C_1 r$$

$$r v_\theta = \frac{C_1 r^2}{2} + C_2$$

$$v_\theta = \frac{C_1}{2} r + \frac{C_2}{r}$$

BC: ① at $r = R_0$, $v_\theta = \omega_0 R_0$

$$\omega_0 R_0 = \frac{C_1}{2} R_0 + \frac{C_2}{R_0} \Rightarrow \frac{\omega_0 R_0^2}{R_i} = \frac{C_1}{2} \frac{R_0^2}{R_i} + \frac{C_2}{R_i}$$

② at $r = R_i$, $v_\theta = \omega_i R_i$

$$\omega_i R_i = \frac{C_1}{2} R_i + \frac{C_2}{R_i}$$

② - ①

$$\omega_i R_i - \frac{\omega_0 R_0^2}{R_i} = \frac{C_1}{2} R_i - \frac{C_1}{2} \frac{R_0^2}{R_i} = \frac{C_1}{2} \left(R_i - \frac{R_0^2}{R_i} \right)$$

$$\Rightarrow \omega_i R_i^2 - \omega_0 R_0^2 = \frac{C_1}{2} (R_i^2 - R_0^2)$$

$$\Rightarrow C_1 = \frac{2(\omega_i R_i^2 - \omega_0 R_0^2)}{R_i^2 - R_0^2}$$

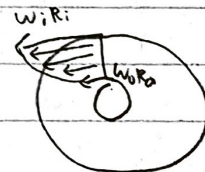
$$\omega_i R_i = \frac{C_1}{2} R_i + \frac{C_2}{R_i}$$

$$\frac{C_1}{R_i} = \omega_i R_i - \frac{C_1}{2} R_i = \omega_i R_i - \frac{(\omega_i R_i^2 - \omega_0 R_0^2)}{R_i^2 - R_0^2} R_i$$

$$C_2 = \omega_i R_i^2 - \frac{(\omega_i R_i^2 - \omega_0 R_0^2)}{R_i^2 - R_0^2} R_i^2$$

$$v_\theta = \frac{C_1}{2} r + \frac{C_2}{r} = \frac{(\omega_i R_i^2 - \omega_0 R_0^2)}{R_i^2 - R_0^2} r + \frac{(\omega_i R_i^2 - \omega_0 R_0^2) R_i^2}{R_i^2 - R_0^2} \frac{1}{r}$$

Velocity profile:



$$3, \quad L = 2 \text{ in} \quad \omega = 200 \text{ rpm} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 20.944 \text{ rad/s}$$

$$\mu = 200 \text{ cp} = 200 \text{ mPa}\cdot\text{s} = 0.2 \text{ Pa}\cdot\text{s} = 0.2 \text{ Pa}\cdot\text{s} \times \frac{2.0886 \times 10^{-2} \text{ lbf}\cdot\text{s}/\text{ft}^2}{1 \text{ Pa}\cdot\text{s}} = 0.004177 \text{ lbf}\cdot\text{s}/\text{ft}^2$$

$$\rho = 50 \text{ lbm}/\text{ft}^3$$

θ - momentum:

$$\rho \left(\cancel{\frac{\partial v_\theta}{\partial t}} + v_r \cancel{\frac{\partial v_\theta}{\partial r}} + v_\theta \frac{\partial v_\theta}{\partial \theta} + v_z \cancel{\frac{\partial v_\theta}{\partial z}} + \cancel{\frac{v_r v_\theta}{r}} \right) = -\cancel{\frac{1}{r} \frac{\partial p}{\partial \theta}} + \mu \left[\cancel{\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right)} + \frac{1}{r} \cancel{\frac{\partial^2 v_\theta}{\partial \theta^2}} + \cancel{\frac{\partial^2 v_\theta}{\partial z^2}} \right. \\ \left. + \cancel{\frac{2}{r} \frac{\partial v_r}{\partial \theta}} \right] + \cancel{\rho v_\theta^2}$$

$$0 = \mu \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right)$$

$$\Rightarrow v_\theta = \frac{c_1}{2} r + \frac{c_2}{r}$$

$$\text{BCs: } (1) \text{ at } r = 1 \text{ in, } v_\theta = \omega r = 20.944 \text{ rad/s} \cdot 1 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}} = 1.7453 \text{ ft/s}$$

$$1.7453 \text{ ft/s} = \frac{c_1}{2} \times 1 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}} + \frac{c_2}{1 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}}} \\ = \frac{c_1}{24} \text{ ft} + \frac{12 c_2}{1 \text{ ft}}$$

$$(2) \text{ at } r = 1.002 \text{ in, } v_\theta = 0 \text{ ft/s}$$

$$0 = \frac{c_1}{2} \times 1.002 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}} + \frac{c_2}{1.002 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}}} \\ = 0.04175 c_1 \text{ ft} + \frac{11.976 c_2}{\text{ft}}$$

$$\Rightarrow c_1 = \frac{-11.976 c_2}{0.04175} \frac{1}{\text{ft}^2}$$

$$1.7453 \text{ ft/s} = \frac{-11.976 c_2}{0.04175} \cdot \frac{1}{24} \cdot \frac{1}{\text{ft}} + \frac{12 c_2}{1 \text{ ft}}$$

$$\Rightarrow c_2 = 36.4331 \text{ ft}^2/\text{s}$$

$$c_1 = \frac{-11.976 c_2}{0.04175} \frac{1}{\text{ft}^2} = -10450.9 \text{ s}^{-1}$$

$$\tau_{r\theta} = -\mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]$$

$$= -\mu r \frac{\partial}{\partial r} \left(\frac{c_1}{2} + \frac{c_2}{r^2} \right)$$

$$= -\mu r \left(-2 \frac{c_2}{r^3} \right) = 2\mu \frac{c_2}{r^2}$$

$$T_z = \tau_{r\theta}|_{r=1 \text{ in}} \cdot 2\pi r L \cdot r$$

$$= 2\mu c_2 \cdot 2\pi L$$

$$= 2 \times 0.004177 \text{ lbf}\cdot\text{s}/\text{ft}^2 \cdot 36.4331 \text{ ft}^2/\text{s} \\ \times 2\pi \times 2 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}}$$

$$= 0.3187 \text{ lbf}\cdot\text{ft} \approx 0.32 \text{ lbf}\cdot\text{ft}$$

$$P = T_z \cdot \omega$$

$$= 0.32 \text{ lbf}\cdot\text{ft} \cdot 20.944 \text{ rad/s} \times \frac{0.001818 \text{ hp}}{1 \text{ lbf}\cdot\text{ft}/\text{s}}$$

$$= 0.012186 \text{ hp} \approx 0.012 \text{ hp}$$

4. $d_1 = 4.00 \text{ cm}$, $d_2 = 4.50 \text{ cm}$, $L = 4.00 \text{ cm}$
 $\mu = 57 \text{ cp} = 57 \text{ mPa}\cdot\text{s} = 0.057 \text{ Pa}\cdot\text{s}$, $\rho = 1.29 \text{ g/cm}^3$

? $T_2 \leq 100 \text{ dyne}\cdot\text{cm} = 0.0001 \text{ N}\cdot\text{m}$

$\omega \leq 0.5\%$

from problem 3, $V_\theta = \frac{C_1}{2}r + \frac{C_2}{r}$

BC1: ① at $r = 2 \times 10^{-2} \text{ m}$, $V_\theta = 0$

$$0 = \frac{C_1}{2} \cdot 0.02 \text{ m} + \frac{C_2}{0.02 \text{ m}}$$

$$= 0.01 C_1 + \frac{50 C_2}{\text{m}}$$

$$\Rightarrow C_1 = -\frac{5000 C_2}{\text{m}^2}$$

② at $r = 2.25 \times 10^{-2} \text{ m}$, $V_\theta = \omega \cdot 2.25 \times 10^{-2} \text{ m} = X \text{ m/s}$

$$X = \frac{C_1}{2} \cdot 0.0225 \text{ m} + \frac{C_2}{0.0225 \text{ m}}$$

$$X = \frac{-5000 C_2}{2 \text{ m}^2} \cdot 0.0225 \text{ m} + \frac{C_2}{0.0225 \text{ m}}$$

$$\Rightarrow C_2 = \frac{X}{-11.8056} \text{ m}^2/\text{s}$$

$$\tau_{r\theta} = -\mu \left[r \frac{\partial}{\partial r} \left(\frac{V_\theta}{r} \right) + \frac{1}{r} \frac{\partial V_r}{\partial \theta} \right]$$

$$= -\mu r \frac{\partial}{\partial r} \left(\frac{C_1}{2} + \frac{C_2}{r^2} \right)$$

$$= -\mu r \left(-2 \frac{C_2}{r^3} \right) = 2\mu r \frac{C_2}{r^3} = \frac{2\mu C_2}{r^2}$$

$$T_2 = \tau_{r\theta}|_r \cdot 2\pi r L \cdot t$$

$$= \frac{2\mu C_2}{r^2} \cdot 2\pi r L = 4\pi \mu L C_2 = -4\pi \mu L \frac{X}{11.8056}$$

$$\Rightarrow X = \frac{11.8056 \cdot T_2}{-4\pi \mu L} = \frac{11.8056 \cdot 0.0001 \text{ N}\cdot\text{m}}{-4\pi \cdot 0.057 \text{ Pa}\cdot\text{s} \cdot 0.04 \text{ m}} = -0.00412 \text{ m/s}$$

$$X = \omega \cdot 2.25 \times 10^{-2} \text{ m}$$

$$\omega = \frac{X}{2.25 \times 10^{-2} \text{ m}} = 0.1831 \text{ rad/s}$$